Proof. First remove the definable locus where the output RV-tuple is not locally constant in the input VF-coordinates. On each resulting open fiber with both input and output RV-tuples fixed, extend the VF-map by zero outside its domain and apply [17, Corollary 9.9] to its partial derivatives. Their nonexistence and discontinuity loci are definable and have VF-dimension less than n in every such fiber. Retaining or projecting away RV-coordinates does not increase VF-dimension, so the combined exceptional locus still has VF-dimension less than n. Off this locus all partial derivatives, and hence their determinant, are continuous. In arity zero the Jacobian is the empty determinant 1. □

Resolved referee R035 [Gap] – Equal VF dimensions and generic RV constancy.
Original source line 993. State $n = n'$ in Lemma 2.38; otherwise the determinant called the Jacobian has not been defined. Moreover, the preceding existence of one full-dimensional fiber for each input t does not show that such fibers cover almost every input point. Use the standard generic local constancy of a definable map from VF^n to $\mathrm{RV}^{m'}$ first, and then apply the cited differentiability theorem on its open fibers. This provides the missing almost-everywhere coverage.

Author resolution. The author approves the square-dimension hypothesis $n = n'$ and the generic local-constancy argument for the output RV-labels. The proof now applies the cited partial-differentiability theorem on open fixed-label charts and explains the smaller-dimensional exceptional locus. The arity-zero Jacobian is the empty determinant 1.

Terminology 2.39. We say that a set $I \subseteq \Gamma_\infty^n$ is *γ-bounded*, where $\gamma \in \Gamma$, if it is contained in the box $[\gamma, \infty]^n$, and is *doubly γ-bounded* if it is contained in the box $[-\gamma, \gamma]^n$. More generally, let A be a subset of $\mathrm{VF}^n \times \mathrm{RV}_\infty^m \times \Gamma_\infty^l$ and

$$\mathrm{valim}(A) = \{(\mathrm{val}(a), \mathrm{vrv}(t), \gamma) \mid (a, t, \gamma) \in A\} \subseteq \Gamma_\infty^{n+m+l},$$

where $\mathrm{val}(a)$ and $\mathrm{vrv}(t)$ are taken coordinatewise. We say that A is *γ-bounded* if $\mathrm{valim}(A)$ is, and likewise for *doubly γ-bounded*. A set is *bounded* if it is γ-bounded for some $\gamma \in \Gamma$, and *doubly bounded* if it is doubly γ-bounded for some $\gamma \in \Gamma$. The notation A_Γ continues to mean projection to existing Γ-coordinates.

Resolved referee R036 [Clarification] – A valuation image is not a coordinate projection.
Original source line 1001. The notation A_Γ here conflicts with Notation 2.1, where it denotes projection to existing Γ-coordinates. Use a separate notation such as $\mathrm{valim}(A)$ for this coordinatewise valuation image. Also explicitly say that 'bounded' means γ-bounded for some γ, and similarly for 'doubly bounded'.

Author resolution. The author approves $\mathrm{valim}(A)$ for the coordinatewise valuation image, reserving A_Γ for projection to existing Γ-coordinates. The existential finite bound in "bounded" and "doubly bounded" is now explicit. This coordinatewise convention is distinct from the minimum tuple valuation used in differentiation.

Definition 2.40 (Contractions)**.** A function $f : A \longrightarrow B$ is rv-*contractible* if there is a (necessarily unique) function $f_\downarrow : \mathrm{rv}(A) \longrightarrow \mathrm{rv}(B)$, called the rv-*contraction* of f, such that

$$(\mathrm{rv} \upharpoonright B) \circ f = f_\downarrow \circ (\mathrm{rv} \upharpoonright A).$$

Similarly, it is res-*contractible* (respectively, val-*contractible*) if the same holds in terms of res (respectively, val or vrv, depending on the coordinates) instead of rv.

Unapplied R037 proof revision for author review. The statement is unchanged. Replace the induction with one relation recording all input and output values. Residue-field/value-group orthogonality gives finite fibers; compactness and the pure value-group structure give finitely many rational-affine branches. A bounded input box then gives one bound for every branch. This supplies the missing

uniformity, rather than assuming fiberwise bounds already bound their union. R036 above is already approved and applied.

Lemma 2.41. *Let $U \subseteq \mathrm{RV}^k$ be a doubly bounded set and $f : U \longrightarrow \mathrm{RV}$ a definable function. Then $f(U)$ is also doubly bounded. The same holds if the codomain of f is Γ.*

Proof. For $f : U \longrightarrow \mathrm{RV}$, set $h = \mathrm{vrv} \circ f$; for $f : U \longrightarrow \Gamma$, set $h = f$. Consider the definable relation

$$R = \{(\alpha, \beta) \in \Gamma^k \times \Gamma \mid \exists u \in U \ (\mathrm{vrv}(u) = \alpha \text{ and } h(u) = \beta)\},$$

where $\mathrm{vrv}(u)$ is coordinatewise. For a fixed α, after choosing representatives of its coordinates, the corresponding RV-fiber is identified with a subset of a product of residue-field multiplicative groups. Orthogonality of the residue field and value group implies that its image under h is finite. By compactness these finite fibers have a uniform bound.

By stable embeddedness, R is definable in the pure divisible ordered abelian group Γ. Its finite fibers therefore lie on finitely many graphs of rational-affine functions of α, on definable pieces of its projection. Since U is doubly bounded, this projection lies in a bounded box in Γ^k. Each of the finitely many rational-affine functions is bounded on that box, and a common finite bound works for all branches. Thus $h(U)$ is doubly bounded. This proves the claim for a Γ-valued f and, by definition of double boundedness, for an RV-valued f as well. The empty set and the case $k = 0$ are included. □

Referee R037 [Gap] – Uniformity in the bounded RV-image lemma.
Original source line 1017. The induction on k must control the bounds uniformly in the remaining coordinates; boundedness on each fiber alone does not give a bound on their union. A precise route is to consider the relation of possible values $\mathrm{vrv}(f(u))$ over $\mathrm{vrv}(u) \in \Gamma^k$. Residue-field/value-group orthogonality makes each such fiber finite, uniformly; stable embeddedness and elimination in divisible ordered abelian groups then give finitely many piecewise rational-affine branches. These have bounded images on a bounded box. Include this justification, or an equivalent explicit uniform induction.

Definition 2.42. Let $A \subseteq \mathrm{VF}^n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ be a definable function whose range is bounded. For $a \in \mathrm{VF}^n$ and $L \subseteq \mathrm{VF}^m$, we say that L is a *limit set of f at a*, written as $\lim_{A \to a} f \subseteq L$, if for every $\epsilon \in \Gamma$ there is a $\delta \in \Gamma$ such that if $c \in \mathfrak{o}(a, \delta) \cap (A \smallsetminus a)$ then $f(c) \subseteq \bigcup_{b \in L'} \mathfrak{o}(b, \epsilon)$ for some $L' \subseteq L$.

A limit set L of f at a is *minimal* if no proper subset of L is a limit set of f at a.

Referee R038 [Clarification; revised] – Simplify the limit-set definition.
Original source line 1046. Say explicitly that $\bigcup_{a \in A} f(a)$ is bounded. The auxiliary subset L' does no work: the displayed condition is equivalent to $f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon)$. The following definition of continuity is applied to finitary functions without a global boundedness hypothesis, so the present definition needs a compatible extension. One can use the same formula for arbitrary correspondences whenever a finite affine limit exists, or formulate limits in $(\mathbb{P}^1(\mathrm{VF}))^m$ from the outset; local boundedness near the point suffices for affine local uses.

Observe that if $\lim_{A \to a} f \subseteq L$ and $b \in L$ is not isolated in L then actually $\lim_{A \to a} f \subseteq L \smallsetminus b$. So in a minimal limit set every element is isolated. Moreover, if a minimal limit set L exists then its topological closure is unique:

Lemma 2.43 ([17, Lemma 9.2])**.** *Let $L_1, L_2 \subseteq \mathrm{VF}^m$ be two minimal limit sets of f at a. Then the closures of L_1, L_2 coincide.*